

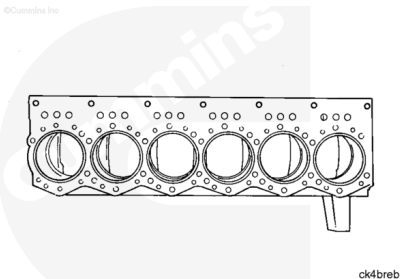
Inspect for Reuse

Check the top surface of the block for wear. If fretting damage is present in an area where a head gasket seal ring or a grommet makes contact, the surface **must** be repaired.

Fretting damage in any other area is acceptable if it does **not** change the protrusion measurement of the liner counterbore.

A newly machined surface **must** be flat within 0.05 mm [0.002 in] under a cylinder head. Waves on the surface are acceptable as long as they are **not** more than 0.018 mm [0.0007 in] high, and the **high** and **low** points of the waves are **not** closer than 25 mm [1.0 in].

A newly machined surface **must** meet the specifications for block height as stated in the next illustration. Reference the following procedure if the top surface of the cylinder liner counterbore **must** be machined. Refer to Procedure 001-058 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-058.html)



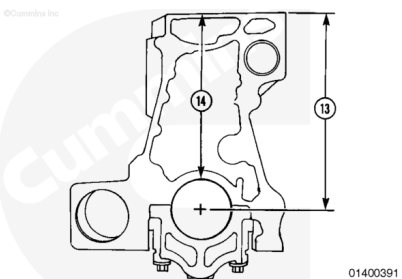
⚠ CAUTION ⚠

The parting line of the main bearing cap is not at the same height as the centerline of the main bearing bore.

Cylinder Block Height (13)		
mm		in
481.91	MIN	18.9735
482.76	MAX	19.0065

If the height of the block is less than 482.42 mm [18.993 in], a 0.51 mm [0.020 in] oversize head gasket **must** be used.

If the checking ring or the centering ring is not available, the height of the block can be measured from the top of the main bearing saddle (14).



Cylinder Block Height - Main Bearing Saddle (14)

mm		in
407.70	MIN	16.051
408.53	MAX	16.084

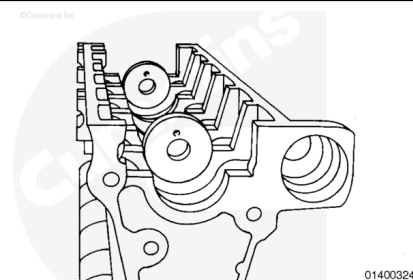
If the block height is less than 408.20 mm [16.071 in], a 0.51 mm [0.020 in] oversize head gasket **must** be used.

The height of the block **must not** vary more than 0.05 mm [0.002 in] from end-to-end of the block. If the block height is **not** within specifications, the top surface of the cylinder block **must** be machined or the block **must** be replaced.

Note : If the top surface of the block is machined, the ledge depth of the cylinder liner counterbore **must** be machined. Refer to Procedure 001-058 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-058.html>)

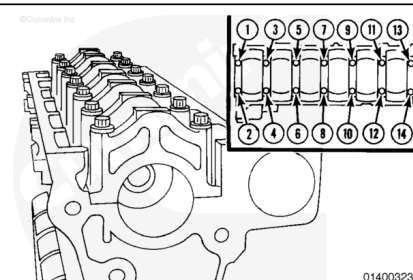
Measure the Main Bearing Bore Alignment

- Use two centering rings, Part Number ST-1177-54 (148.5 mm [5.846 in] diameter). Install the rings in the Number 2 and Number 6 main bearing locations.



⚠ CAUTION ⚠

The main bearing caps are numbered 1 through 7. Install each cap in the correct location. The slot in the cap for the bearing tang must be on the same side as the slot in the cylinder block.



Install each capscrew by hand to make sure the capscrew is positioned properly.

Use a mallet to install the main bearing caps until they touch the cylinder block.

If any of the main bearing caps do not require force during installation, mark the cap to check the side clearance.

Tighten the main bearing caps using two steps:

Torque Value:

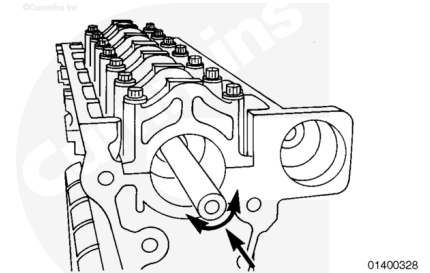
1. 265 n•m [196 ft-lb]

2. 605 n•m [448 ft-lb]

Note : This torque sequence is for testing purposes only. Reference the following procedure for installation torque specifications. Refer to Procedure 001-006 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-006-tr.html>)

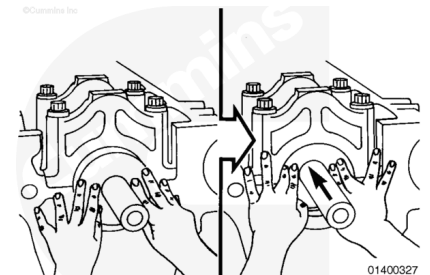
Use lubricating engine oil to lubricate the inside diameter of the two centering rings.

Install the alignment/boring bar, Part Number ST-1177-16, in the two centering rings. The bar **must** turn easily. If the bar does **not** turn easily, check to make sure that the main bearing caps are installed correctly. If the bearing caps appear to be OK, move one of the centering rings to another bearing location.



Install the checking ring in the main bearing bore by hand. If the ring will **not** slide through the bore, check the bore burrs. If the ring will still **not** slide through the bore, the bore is undersize and **must** be repaired.

Reference the Alternative Repair Manual, Bulletin 3379035.



Use a 0.075 mm [0.0003 in] feeler gauge that is **not** more than 13 mm [0.5 in] wide.

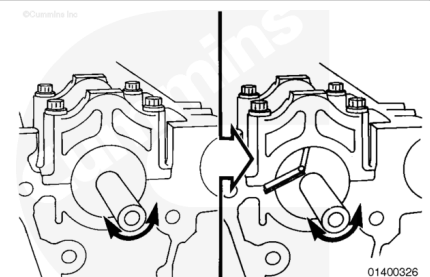
Center the checking ring in the bore. Try to put the feeler gauge between the checking ring and the bore. Rotate the gauge in the bore at both sides of the checking ring.

The bore alignment of the main bearing is OK if:

- The gauge does **not** enter at any point
- The gauge will enter, but will **not** slide through or around the bore. The alignment bar will rotate with the gauge inserted.

The bore alignment of the main bearing is **not** OK if:

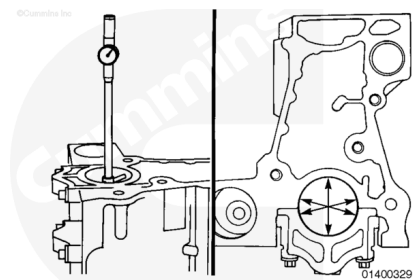
- the gauge enters and slides around the bore. This means that the bore is oversize and **must** be repaired.
- the gauge will enter on one side **only**, but can slide around the bore. This means that the bore is tapered and **must** be repaired.



If the tools to check the main bearing bore alignment are **not** available, use a dial bore indicator.

Measure the inside diameter in the positions shown. This inside **must** be completely round within 0.013 mm [0.0005 in].

Note : Support the rear portion of the block on the flat surface to obtain the most accurate measurement of inside diameter.

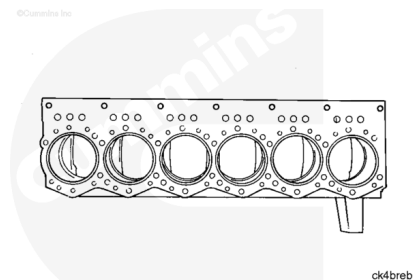


Main Bearing Bore I.D. (Capscrews Torqued to Specification)

mm		in
148.45	MIN	5.8444
148.50	MAX	5.8465

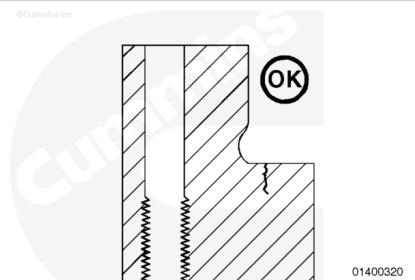
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.

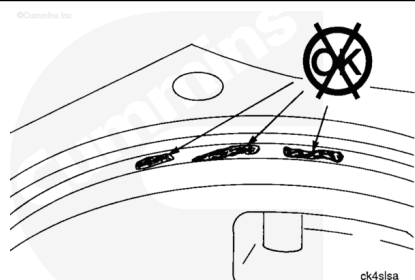


Clean the counterbore area with safety solvent.

Metallurgical analysis of cross sections of counterbores having circumferential cracks has revealed that the cracks are surface-initiated on the top of the counterbore ledge, and normally do **not** propagate vertically through the counterbore ledge into the coolant passage around the cylinder liner.



Pitting on the liner seat is **not** acceptable. This picture illustrates an example of pitting in a damaged area. This block requires counterbore machining before it can be reused.

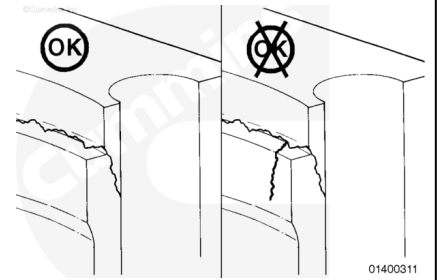


Magnetic Crack Inspect

Use crack detection kit, Part Number 3375432, or equivalent, to check the counterbore ledge for cracks.

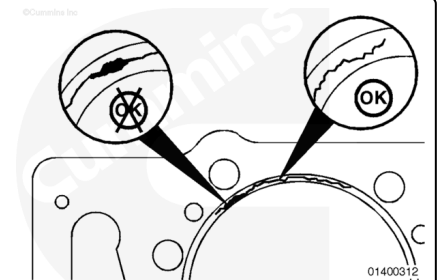
Circumferential cracks of the counterbore ledge are acceptable if the cracks do **not** extend to or over the edge of the ledge as shown in the illustration.

Circumferential cracks in the radius are acceptable if they do **not** extend more than 90 degrees around the circumference of the counterbore radius.



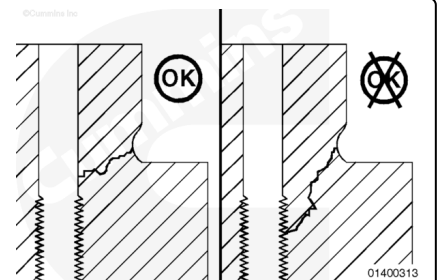
It is **not** necessary to machine the block in an effort to remove acceptable cracks. If cracks that are **not** acceptable are found during the initial inspection, the counterbore **must** be machined. Contact a Cummins Authorized Repair Location.

If a crack that is **not** acceptable remains after the machining repair procedure is completed, the block is **not** acceptable for use.



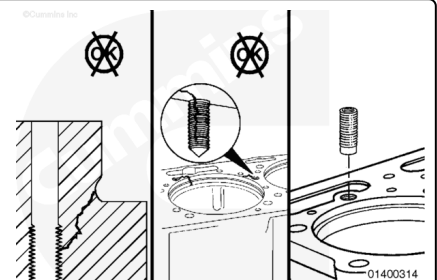
Check the capscrew holes for cracks.

Cracks that extend from the counterbore wall to the capscrew hole are acceptable for use **only** if they do **not** extend into the threaded portion of the hole.



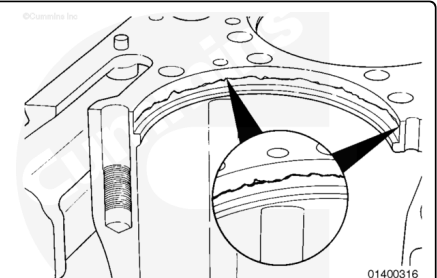
Cracks that extend into the threaded portion of the capscrew hole require repair with a blind-end thread insert.

Contact an Authorized Cummins Repair Facility for the proper threaded inserts.



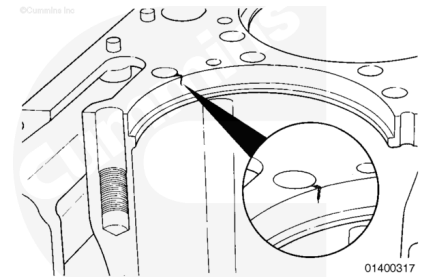
Check for cracks running horizontally around the vertical wall of the counterbore.

All coolant passages that are closest to the bore **must** be repaired with coolant passage threaded inserts.



Check for cracks that run vertically to a coolant passage or a capscrew hole.

Those passages **must** be repaired with coolant passage threaded inserts.



Cylinder blocks with vertical cracks that extend from a coolant passage down over the counterbore ledge can **not** be repaired.

